

ROUNAK RAMAN

+918826879389 | Email | [Linkedin](#) | [Github](#) | [Google Scholar](#) | [Orchid](#) | [Publons](#) | [Portfolio Website](#) | [Substack](#) | [Medium](#) | [LeetCode](#) | [Linktree](#)

RESEARCH SUMMARY

Researcher focused on secure, trust-aware, and energy-efficient protocols for IoT, wireless sensor, and opportunistic networks, with **four peer-reviewed publications (3 SCI-indexed, including a first-authored Q1 paper in *Internet of Things*, Elsevier)** spanning **context-aware data aggregation, hierarchical network security, and swarm-optimization-driven cluster management**. Parallel applied research experience in **AI-driven decision systems** (manufacturing HR analytics), **public policy** (innovation/startup ecosystems at NITI Aayog), and **quantitative finance** (STIR derivatives research at Futures First), reflecting a broader interest in applying rigorous, data-driven methods across technical and applied-policy domains.

EDUCATION

Netaji Subhas University of Technology (Erstwhile NSIT), Delhi, India

Aug' 21-Aug'25

Bachelor of Technology in Information Technology (CVSPK Scholar/Department Rank 1)

CGPA=8.95/10

Kulachi Hansraj Model School, Delhi, India

Apr'20 -Apr'21

Class XII (High School): Central Board of Secondary Education

Percentage = 95.2%

SKILLS & RESEARCH INTERESTS

Programming Languages: C, C++ (memory management, and data structures), Python (NumPy, pandas, SciPy, Matplotlib, Seaborn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-Learn, and OpenCV), SQL & NoSQL (MySQL, PostgreSQL, MongoDB), Shell Scripting (Bash), HTML, CSS, Javascript (Backend development and API handling using Postman API), MATLAB
Frameworks & Libraries: Machine Learning & Deep Learning (**Supervised Learning:** Regression, classification, and ensemble methods like XGBoost, LightGBM, CatBoost, and Scikit-Learn, **Unsupervised Learning:** Clustering algorithms (K-Means), dimensionality reduction techniques (PCA, t-SNE), and anomaly detection, **Deep Learning:** FastAI, CNNs, RNNs, transformers, and large language models (LLMs), Model Training, Tuning & OpenAI Models (Hugging Face Transformers and Gemini, Whisper, GPT, Dall-E)), **Natural Language Processing (NLP)** (Hugging Face Transformers, spaCy, NLTK, BERT, ChatGPT), **Computer Vision** (OpenCV, YOLO, OCR) , Docker, Seaborn, Matplotlib, Plotly, Power BI, Omnet++, INET, Latex, Git and GitHub

Core Competencies: DSA, OOPs, OS, DBMS, Computer Networks, Cryptography

Areas of Interest: Generative AI (GenAI), Natural Language Processing (NLP), Edge AI and IoT, Cyber Security & Cryptography, Financial Technology (FinTech), Data Analysis & Predictive Modelling, Wireless Sensor Networks & Opportunistic Networks, AI-Driven Network Management, Self-Healing IoT Networks and Fog Computing

EXPERIENCES

Financial Market Analyst - Money Market & Interest Rates Researcher | Futures First | [LOR](#) **Jan 2025–Present**

Proprietary trading house specializing in fixed income & interest rate derivatives; full P&L ownership across global STIR markets

- Conducted applied research on U.S. money markets, analyzing **SR3/SR1 and SR3/ZQ** spread dynamics to identify relative-value opportunities across SOFR and Fed Funds futures; own **full P&L on a live US \$500,000 portfolio**
- Researched U.S. Treasury-repo market** microstructure using SOFR–EFFR futures contracts and Treasury swap spreads, conducted **macro research on GSIB balance sheets** across 3+ capital, liquidity and funding scenarios to inform desk-level positioning
- Designed and back-tested multi-factor relative-value research strategies (basis trades, steepener/flattener structures, multi-leg butterflies) with transaction-cost modeling, walk-forward validation and out-of-sample stress testing across 8+ strategy configurations
- Documented research findings via automated intraday research reports (**realized vs. modeled slippage, margin burn, roll cost**) and desk-level execution SOPs, reducing P&L leakage across the desk
- Ranked **Top 20% of 400+ interns**, awarded **Pre-Placement Offer (PPO)** and confirmed to full-time role following probation

Research Intern – First-Author Manufacturing AI Study | IIM Indore | [Report](#)

Mar'26-Apr'26

Under [Prof. Nishit Kumar Sinha](#), manuscript finalisation and publication preparation.

- Finalized the 27-page first-author manuscript “From Hiring to Safety: Practical AI Use Cases in Manufacturing HR,”** refining the argument, section architecture, figures, tables, references and publication-ready appendix.
- Converted a 2015–2025 evidence base and 32-source bibliography into a sector-specific framework covering AI-assisted hiring, attrition prediction, rostering, learning, performance analytics, PPE/fatigue monitoring and workforce planning.
- Developed a 7-domain AI readiness checklist and implementation toolkit with 8–12 week pilots**, baseline KPIs, ROI gates, edge-first architecture, open-API integration, bias/fairness testing, DPDP/GDPR alignment and human-in-the-loop oversight

Financial Research Consultant – Innovation & Startup Ecosystems | NITI Aayog, Government of India **Nov'25-Jan'26**

India's apex public policy think-tank driving economic strategy, investment frameworks & innovation ecosystems

- Conducted comparative research across 15+ innovation and startup-policy frameworks**, evaluating funding mechanisms, academic participation, commercialization pathways and startup outcomes to identify structural gaps in capital deployment.
- Cleaned, structured and quality-checked 500+ Atal New India Challenge records**, strengthening data integrity and traceability for government disbursement and programme monitoring.
- Synthesized **India's US\$140B+ startup ecosystem**, venture-funding trends and **sectoral investment flows into 3+ analytical reports** and evidence-based policy recommendations, received a supervisor [letter of recommendation](#).

Research Intern - Winter (First Author) | NPTEL / Anna University | [LOR](#)

Dec'25–Jan'26

Under Dr. Chitra Dey; initial research design and manuscript development.

- Designed the initial AI-in-manufacturing-HRM study, literature-search strategy and evidence matrix using Scopus**, Web of Science and Google Scholar queries spanning 2015–2025.
- Mapped 10+ use cases** across talent acquisition, predictive attrition, learning, performance management, skill forecasting, workforce scheduling and safety/compliance; defined practical data inputs, pilot metrics and adoption barriers.

- **Drafted the AI-HRM readiness and governance framework** covering data infrastructure, digital literacy, worker communication, ethics/legal controls, cross-functional governance, model feedback and continuous ROI measurement.
- **Performed quantitative analysis and visualisation for an associated gig-economy study**, including employment projections, sector classification and skill-mix trends.

Research Intern - Summer | NPTEL / Anna University | [\[LOR\]](#)

May'25-Jun'25

Selected through the NPTEL topper internship pathway under Dr. Chitra Dey.

- **Executed a PRISMA-compliant systematic review** on gig-platform fairness; built advanced Boolean search strings across Scopus, Web of Science and IEEE Xplore, **screening 150+ records to retain 45 high-relevance studies**.
- **Synthesized 30+ AI-in-HR papers**; distinguished reported study outcomes such as up to **70% lower manual screening time** and **30% higher training completion** from implementation risks including bias, privacy and change resistance.
- Built reproducible research infrastructure comprising bibliographic pipelines, Excel-based inclusion/exclusion audit trails, coded evidence matrices and thematic-analysis frameworks.
- **Authored and presented a 60-page report on digital-platform governance and ethical AI adoption** for workshop submission and future publication development.

Research Intern - Consumer Analytics, Digital Marketing & Brand Ethics | IIM Kozhikode | [\[LOR\]](#)

Mar'25-May'25

- **Compared 8 global creators with audiences of 1.5M–16M**, identifying three cross-market content strategies and translating findings into international brand-campaign hypotheses.
- **Co-authored a 30-page branded-hashtag study** including #LanguageLearning (5,121 posts in six months) and #ripduo (45,000+ uses; 25,560% spike in mentions), with recommendations for reach, timing and narrative design.
- **Analysed 500+ social mentions of Facebook's transition to Meta**, finding a **2:1 neutral/negative-to-positive response ratio** and that 48% of users prioritised brand responsiveness; developed implications for rebranding communication.
- **Coded 200+ Reddit/X threads on brand moral failures** across India and the U.S. and **analysed a 500+ respondent OTT dataset**; developed a care–compliance–competency framework, predictive purchase-intent model and Excel scenario dashboard.

R&D Intern | DRDO-INMAS | Ministry of Defense, Government of India | [\[LOR\]](#) | [\[Project Report\]](#)

Jun'23-Aug'23

Research Intern under Rajesh Kumar Tiwari, Scientist 'F' in DRDO-INMAS CCML Lab

[Offer Letter](#)

- **Developed an EEG Data Analysis System** using Neuromore Studio, enabling real-time brainwave visualization and conducting **20+ neurofeedback sessions** that improved **attentiveness by 15-20%**.
- **Engineered a multimodal neurofeedback loop** with visual (candle brightness) and auditory (music volume) cues, **achieving 25% higher subject engagement** and a **20% reduction in session time**.
- **Integrated a predictive neural network with 85% accuracy** to assess cognitive states pre-session, **cutting setup time by 30%** and streamlining subject progress tracking.
- **Utilized Fast Fourier Transform (FFT)** to transform EEG data into the frequency domain, enabling calculation of **averageAlpha wave amplitude** for providing targeted feedback to subjects.
- Explored **EEG applications for personalized military training and public health**, highlighting potential for enhanced cognitive performance assessment and targeted interventions.

Research Intern | Department of Information Technology | NSUT | [\[LOR\]](#) | [\[Project Report\]](#)

Aug'23-Dec'23

Research Intern under [Dr. Deepika Kukreja](#) and Ms. Richa Garg, Information Technology Dept, Delhi, NSUT

[Offer Letter](#)

- **Designed and implemented EAHCP** (Energy Aware Hybrid Clustering Protocol) for dynamic IoT networks using **MATLAB and Simulink**, achieving **20% improvement in energy efficiency** and **35% increase in network lifetime** compared to conventional protocols like SEP, Z-SEP, and ETM-IoT. Protocol tested in environments with up to **200 nodes**, leveraging hybrid clustering for **both static and mobile nodes**.
- **Enhanced message reliability** by over **25% using node-based routing** and **convex hull boundary algorithms** to maintain network topology within a 100m x 100m region, demonstrating a **40% reduction in dead nodes under high-mobility**.
- **Optimized performance metrics** (energy consumption, throughput, and packet delivery) through simulation across diverse network topologies in Simulink, with residual energy evaluations ensuring prolonged network stability and high data fidelity across **80% mobile nodes**.
- **Exposure:** IoT Network Design, MATLAB, Simulink, Network Optimization, Convex Hull Algorithms and Latex

PROJECTS

PRAGATI – Traffic Analyser and Parking Spot Detector | [\[CODE\]](#) | [\[Working Prototype\]](#)

Nov'23

Smart India Hackathon, Ministry of Housing & Urban Affairs, Government of India

[Idea Report](#)

- **Developed Pragati**, an innovative smart parking solution that **leverages YOLO (You Only Look Once) object detection technology** to **accurately identify and monitor available parking spaces in real-time**, enhancing urban mobility.
- **Created a user-friendly web application** that provides real-time parking availability information, **enabling users to quickly locate and reserve parking spaces, reducing time spent searching for spots by up to 30%**.
- **Focused on social impact**, the project aims to alleviate the stress of urban parking issues while **promoting sustainable transportation practices**, contributing positively to community well-being and environmental preservation.
- **Secured Runner-Up position at the SMART INDIA HACKATHON**, competing against **500 teams**, which underscored the project's effectiveness and potential impact in addressing urban parking challenges.
- **Selected as a Millennium Fellow for the Class of 2024**, recognized as **one of 4000 students out of 200,000 applicants**, reflecting a commitment to social change and leadership in developing solutions like PRAGATI **targeted for SDG 11**.
- **Exposure:** Object Detection (YOLO), Web Application Development, Urban Mobility Solutions, Smart Contracts and OAuth.

SyntheX - Advanced Generative AI Model for Internal Document Analysis | [\[CODE\]](#) | [\[Prototype\]](#)

Sep'24

Accenture Innovation Challenge, Organized by Hack2Skill

[Prototype Report](#)

- **Led the development of SyntheX**, an advanced generative AI model designed for organizations' internal document analysis, facilitating the simultaneous examination of multiple documents and **enhancing information retrieval efficiency by over 40%**.
- **Implemented FAISS AI semantic search model**, enabling effective querying of vast document repositories, which **improved**

response time for document searches by 50% compared to traditional methods, streamlining organizational workflows.

- **Integrated Whisper model to transcribe voice notes and call recordings** from meetings or vendor discussions into text-based transcripts, enhancing documentation accuracy and **reducing manual entry time by 60%**.
- **Transformed transcripts into smart notes**, allowing users to generate actionable insights and summaries, thus improving knowledge management and accessibility across the organization.
- **Achieved Top 10 position out of 4,000 registered teams in Accenture Innovation Challenge**, showcasing the effectiveness and innovation of the SyntheX model in real-world applications.
- **Exposure:** Generative AI, Natural Language Processing (NLP), Semantic Search, Whisper Model, Document Analysis, Advanced Algorithms, Real-time Data Processing and Docker.

Meesho QuickTag - AI-Powered Product Attribute Identification Tool

Oct'24

Meesho Visual Taxonomy Challenge, Meesho- India's 3rd largest E-Commerce Startup

- **Spearheaded** the development of F.R.I.D.A.Y, an advanced system for real-time facial recognition, leveraging computer vision techniques and deep learning models.
- **Developed the Meesho QuickTag tool** utilizing the **microsoft/Phi-3.5-vision-instruct model**, implementing **AutoModelForCausalLM** for processing and analyzing product images. **Leveraged CUDA for optimized performance** in model inference, ensuring efficient computation on GPU resources.
- **Conducted extensive preprocessing** on a dataset of **70,000 images**, visualizing data and removing null values to ensure data quality. Successfully addressed **multi-label attribute classification problem**, model predicted attributes for product images.
- **Reduced manual tagging time** by approximately **60%**, allowing teams to focus on higher-value tasks. Achieved a **90%+ accuracy rate** in attribute classification, **improving customer satisfaction via better product searchability and relevance**.
- **Exposure:** AI Model Development, Data Preprocessing, Multi-Label Classification, Image Processing, Front-End and Back-End Development, Advanced Natural Language Processing Techniques.

PUBLICATIONS

HKRISRP: Hierarchical Key Rotation and Isolation Protocol for Secure Wireless Sensor Networks | [\[VIDEO\]](#) Jun'26

Intl. Journal of Ad Hoc and Ubiquitous Computing, Inderscience - Vol. 52, 2026 | Scopus | IF: 1.8 [DOI: 10.1504/IJAHUC.2026.153826](#), under the guidance of [Dr. Nisha Kandhoul](#), [Dr. S.K. Dhurander](#), and [Dr. Isaac Woungang](#)

- **Developed the Hierarchical Key Rotation and Isolation Protocol (HKRISRP)**, enhancing security, robustness, and **energy efficiency** for IoT applications in critical sectors like **military and border surveillance**.
- **First-authored** a hierarchical security protocol combining dynamic key rotation, lightweight encryption, and adaptive cluster-head re-election to defend WSNs against internal/external compromise
- Built **SURAKSHA (Sensor-based Unmanned Resilient Anti-border Knowledge Security Hub Architecture)**, an **Arduino/NRF hardware prototype**, to validate real-time malicious-node detection and network recovery
- Achieved **~10,000 bps throughput**, **82% average residual node energy**, and **0.8s mean threat-isolation time** across OMNeT++ and hardware trials
- **Implemented dynamic encryption with hierarchical key rotation** and lightweight cryptography, **achieving 20% greater network resilience** and **30% better energy efficiency** than traditional protocols in Omnet++ simulations.
- **HKRISRP reduced mean-time-to-isolation by 50%**, improved network **throughput by 15%** over comparable protocols.

Research Exposure: Wireless Sensor Networks, IoT Security, Hierarchical Network Architecture, Dynamic Key Rotation, Lightweight Cryptography, Malicious-Node Isolation, Cluster-Head Election, Arduino, NRF Modules, OMNeT++, INET Framework and Hardware Prototyping.

EESCM: Energy-Efficient and Secure Cluster Management via Convex-Hull-Driven SSA-WWOA

Jun'26

Cluster Computing, Springer Nature - Vol. 29, Jun 2026 | SCIE, Scopus Q1 | IF: 4.1 | [DOI: 10.1007/s10586-026-06073-1](#)

- Co-developed a hybrid optimization framework (Salp Swarm + Wagging-Based Whale Optimization + Convex Hull) for secure, energy-aware cluster-head selection in IoT networks
- Jointly optimized cluster-head selection across residual energy, node degree, trust level, and coverage, improving decision consistency via statistical parameter tuning
- Outperformed baseline clustering algorithms with **20.7% fewer dead nodes**, **14.2% higher packet-delivery ratio**, **4.8% higher throughput**, and **17.2% greater residual network energy**

Research Exposure: Internet of Things, Energy-Efficient Clustering, Secure Cluster Management, Salp Swarm Algorithm, Whale Optimization Algorithm, Convex Hull, Cluster-Head Selection, Trust-Aware Networking, Metaheuristic Optimization and Network Performance Evaluation.

CONTEXT-NET: Context-Aware Nexus-Based Aggregation Protocol for Opportunistic Networks

Nov'25

Internet of Things, Elsevier | Volume 34, Article 101809 | SCIE, Scopus Q1 | IF: 6.0 | SJR: 1.527 | [DOI: 10.1016/j.iot.2025.101809](#)

- **First-authored** a context-aware aggregation protocol for opportunistic networks, integrating a nexus-ring topology, autoencoder-based compression, and ACO-inspired adaptive routing
- Engineered a multidimensional context-evaluation model (spatial, temporal, environmental) to drive real-time transmission prioritization
- Validated via MATLAB simulations, delivering **94.72% throughput stability**, **6.45% higher delivery probability**, and **28% lower one-hop delay** as network scaled **50→100 nodes**
- Cut end-to-end delay by **7.9%** and communication overhead by **5.96%**, demonstrating scalable performance under intermittent connectivity

Research Exposure: Opportunistic Networks, Context-Aware Computing, Data Aggregation, Autoencoders, Ant Colony Optimization, Adaptive Routing, Collision Avoidance, Trust Management, MATLAB and Network Performance Analysis.

ARMor-IoT: Aggregated Reliable Mechanism for Optimized Trust in IoT

Nov'25

ICAIA 2025, Springer Nature | Scopus Indexed | [DOI: 10.1007/978-981-95-0493-0_18](#)

- **First-authored** a multidimensional trust-scoring framework (direct, indirect, core trust) to detect and exclude malicious nodes from IoT data aggregation
- Applied TOPSIS multi-criteria ranking to classify nodes as trusted/malicious with rapid, adaptive decisioning

- Demonstrated high data-delivery success with low latency via simulation, establishing a scalable alternative to centralized authentication in heterogeneous IoT networks
- Research Exposure:** IoT Security, Trust Management, Direct and Indirect Trust Modelling, Multi-Criteria Decision-Making, TOPSIS, Malicious-Node Detection, Secure Data Aggregation, Behavioural Assessment and Network Reliability.

ACADEMIC SERVICE & PEER REVIEW

- **Editorial Board Member**, *International Journal of Sensors and Sensor Networks* | **Jul'26–Present**
- **Invited Peer Reviewer**, *Journal of Field Robotics*, Wiley | **Jan'26–Present**
- **Peer Reviewer**, *International Journal of Reconfigurable and Embedded Systems* | **Jan'26–Present**
- **Peer Reviewer**, *Journal of Engineering Research and Sciences* | **Feb'26–Present**
- **Conference Reviewer**, IEEE ICECCME 2026 | **Apr–May'26**

RELEVANT COURSEWORK

- **NISM Series XA** – Investment Advisor Certificate **107 / 150 (Exp. Apr 2029)**
- **NISM Series VA** – Mutual Fund Distributors Certificate **82 / 100 (Exp. Mar 2029)**
- **NISM Series VIII** – Equity Derivatives Certificate **85 / 100 (Exp. Feb 2029)**
- **NISM Series XV** – Research Analyst Certificate **(Exp. Sep 2028)**
- **AMFI Registered Mutual Fund Distributor (ARN-354994)**
- **Data Analytics with Python** (NPTEL, IIT Roorkee)
- **Ethical Hacking** (NPTEL, IIT Kharagpur)
- **Cyber Defense Engineering using Python** (CyberVidyaPeeth)
- **Python for Data Science and Machine Learning** (Jose Portilla, Udemy)
- **Deep Learning (Andrew Ng, Coursera)**
- **System Design & Architecture (Coding Ninjas)**
- **MySQL (Colt Steele, Udemy)**
- **CS50x: Computer Science (Harvard University)**
- **Web Development (Code With Harry, IIT Kharagpur)**

POSITIONS OF RESPONSIBILITY

Founding Office President-GovTech Office (NSUT) | Initiative by Samagra Consulting **Feb'24–Present**

- Spearheaded the establishment of a Gov-Tech office within a month, recruiting 50+ contributors for DPG/DPI development for SamagraX and ONDC platform. Taught students Git/Github and front-end development
- Conducted outreach to 200+ students through workshops, facilitated research collaborations between students and faculty, resulting in two research groups producing academic papers.

Vice-President | NAKSHATRA-The Astronomical Society of NSUT **Jul'23-Jul'24**

- Led Triathlon event as POC, managing 150+ participants, driving 80k+ social media impressions, securing 10+ sponsors (GAIL, Unacademy), and contributing to record-breaking Resonanz event with over 1000+ registrations.
- Managed Nakshatra Instagram account, increasing follower count by 500, fostered industry connections, and won **Quarka Astro Quiz at IIT Dhanbad**, enhancing society's external presence.
- Authored solo blog, "**Jupiter Via A New Lens**", featured on society's Instagram and Astro Journal, demonstrating academic and writing abilities.

EXTRA CURRICULAR ACTIVITIES

Mentor | GirlScript Summer of Code **Sep'24 - Present**

Managed open-source projects by creating issues, assigning labels, and categorizing by difficulty to facilitate participation. Mentored contributors with technical guidance, pull request reviews, and constructive feedback to improve code quality.

Content Writer | TEDxNSUT **Mar'23 - Apr'23**

Curated content for notable speakers like Amogh Lila Prabhu and Sahil Pruthi for TEDxNSUT's "Painting Portraits" event. Cold-mailed speakers, including Prajakta Kohli and Dr. Vikas Divyakirti, inviting them as guest speakers. Oversaw the registration desk to ensure a seamless experience for more than 2k attendees.

HONOURS AND ACHIEVEMENTS [\[LINK\]](#)

- Secured a rank of **7851 out of 1.5 million students** candidates in JEE ADVANCED 2021. Secured seats in IITs, ISM & IIST
- Awarded the CVSPK Scholarship as the **Department Rank 1** for the **2023-24 & 2024-25** academic year.
- **Four-Time Best Research Award**, Netaji Subhas University of Technology - Recognised for exceptional academic performance and peer-reviewed publications in high-impact journals.
- **InRes Young Researcher Award 2025**, Institute of Researchers - Nov 2025 and **Young Scientist Award** - Oct 2025.
- **GATE Computer Science 2025 Qualified**, IIT Roorkee - Qualified in GEN category on the first attempt among 170,000+ students.
- Recipient of the award for **best presentation at the Delhi Productivity Council Conference** on "AI for Economy"
- **Placed 2nd among 400+ teams** in the **Astronomy Case Study Competition, Astrax'22, IIT Mandi.**
- **Top 5 in WIE Ideathon** by IEEE GGSIPU for our project named **Rentability**-an E-Commerce app for college students.
- **First Position in 'Design in Collaboration using Geometry'** on GANIT WEEK.
- **NEC finalist**, an all-India Entrepreneurship competition conducted by IIT Bombay. Interacted with VCs & investors.
- **Pitched fintech startup 'PayLYT'** at NSUT-Incubation Centre, securing **second place** and receiving **seed funding** for
- Selected as an **International Delegate** representing India among 500 global delegates at **HPAIR '24 in Bangkok.**
- **5-star SQL problem solver** on HackerRank demonstrating strong proficiency in SQL and query writing.
- **All-India Topper** for Data Analysis Using Python Course by IIT Roorkee, scoring over 90% in the final test.